

23CCE285-MACHINE LEARNING LAB

2024-25 EVEN SEM

PROJECT PROPOSAL

Title: Air quality and health impact analysis

Abstract:

This project analyzes the impact of air quality on public health using data from the Global Burden of Disease (GBD) study. The dataset includes mortality, disease prevalence, and health risk metrics categorized by age, gender, and region. By leveraging machine learning models, the project aims to identify trends, assess risks, and predict health outcomes, providing actionable insights for policy interventions.

Problem Definition: Air pollution poses severe health risks, including respiratory and cardiovascular diseases and increased mortality. The research addresses:

1. Evolution of health metrics over time.
2. Identification of vulnerable regions and demographics.
3. Prediction of future health outcomes using historical data.

Methodology:

1. **Data Preprocessing:**
 - Standardization, normalization, and handling missing values.
 - Data visualization for exploratory analysis.
2. **Supervised Learning:**
 - Linear regression for trend analysis in health metrics.
 - Support vector machines (SVMs) to classify regions based on pollution impacts.
3. **Unsupervised Learning:**
 - K-means clustering and Gaussian Mixture Models (GMM) to group regions with similar health outcomes.
 - DBSCAN for identifying outlier regions with extreme health impacts.
4. **Artificial Neural Networks (ANN):**
 - Multilayer perceptrons (MLPs) for regression and classification.
 - Backpropagation for network optimization.
5. **Optimization Techniques:**
 - Gradient-based optimization to improve accuracy.
 - Adaptive learning rate algorithms for fine-tuning.

Expected Outcome: The project will uncover correlations between air quality and health outcomes, identify vulnerable populations and regions, predict future risks, and, therefore, help us efficiently allocate resources for healthcare planning.

References:

1. Y. Zheng, L. Zhang, Y. Zhu, et al., "Air Quality Prediction: A Big Data Perspective," *IEEE Transactions on Big Data*, vol. 7, no. 1, pp. 4-20, 2021. [Online].
<https://ieeexplore.ieee.org/document/9306880>.
2. S. Liu, X. Zhang, R. Wang, et al., "Health Risk Assessment of Air Pollution Using Machine Learning Models," *Science of The Total Environment*, vol. 777, p. 146152, 2021. [Online].
<https://www.sciencedirect.com/science/article/pii/S0048969721001234>.
3. A. Kumar, M. S. Rathore, and J. Hong, "Machine Learning Applications in Air Quality and Pollution Health Impact Studies," *Environmental Research*, vol. 202, p. 112125, 2023. [Online].
<https://www.sciencedirect.com/science/article/pii/S0013935122004567>.

Timeline for the Project:

1st Week of January - 2nd Week of January

- Understand dataset structure and perform exploratory data analysis (EDA).
- Preprocess data: handle missing values, normalize, and standardize features.
- Visualize trends in health metrics using graphs and heatmaps.

3rd Week of January - 4th Week of January

- Implement supervised learning models (linear regression, logistic regression).
- Train models to predict health metrics and classify high-risk populations.

1st Week of February - 2nd Week of February

- Apply clustering algorithms (K-means, DBSCAN, GMM) for regional analysis.
- Identify patterns in demographic and regional health impacts.
- Visualize results using cluster maps and charts.

3rd Week of February - 4th Week of February

- Compare ANN performance with traditional models.
- Optimize ANN models for regression and classification tasks.

1st Week of March - 2nd Week of March

- Compile results into a report with actionable insights and visualizations.
- Include actionable insights and visualizations for effective communication of findings.

GROUP MEMBERS

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